GOKARAJU RANGARAJU INSTITUTE OF ENGINEERING AND TECHNOLOGY

AUTOMATA AND COMPILER DESIGN

Course code:GR22A3115 **L/T/P/C: 2/1/0/3**

III Year II Semester

Prerequisites:

Students should have Knowledge in Data Structures, Simple graph algorithms, Programming language.

Course Outcomes:

1. Articulate the models for regular sets grammar for program constructs.
2. Develop the phases of the compiler for a language definition through regular expressionsand grammars.
3. Build the Bottom up parsers and intermediate code generation algorithms.
4. Experiment memory management techniques in the phases of compiler.
5. Interpret the Code optimization and code generation techniques.

UNIT I

Introduction: Alphabets, Strings and Languages; Automata and Grammars, Deterministic Finite Automata (DFA)-Formal Definition, Simplified notation: State transition graph, Transition table, Language of DFA, Nondeterministic Finite Automata (NFA), introduction to CFG, Ambiguous grammar, Regular Expressions, Arden's theorem.

UNIT II

Compiler Structure: Compilers and Translators, Various Phases of Compiler, Pass Structure of Compiler, Bootstrapping of Compiler. **Lexical Analysis**: The role of Lexical Analyzer, The Lexical- Analyzer Generator Lex , **Syntax Analysis** : CFG, Lexical Versus Syntactic Analysis , Eliminating Ambiguity , Elimination of Left Recursion , Left Factoring, **Top-Down parsers**: Recursive-Descent Parsing, FIRST and FOLLOW, LL(1) Grammars, Non recursive Predictive Parsing, Error Recoveryin Predictive Parsing.

UNIT III

Bottom–up Parsers: Reductions, Handle Pruning, Shift-Reduce Parsing, Conflicts During Shift- Reduce Parsing, LR parsers : SLR, Canonical LR, LALR, Parser Generators : YACC. Syntax Directed translation mechanism and attributed definition.

Intermediate Code Generation: Variants of Syntax Trees, Three-Address Code, Quadruples & Triples, Types and Declarations, Translation of Expressions, Type Checking, Control Flow, Switch- Statements, Intermediate Code for Procedures.

UNIT IV

Run Time Memory Management: Static and Dynamic storage allocation, stack based memory allocation schemes, Symbol Table management, Error Detection and Recovery: Lexical phase errors, Syntactic phase errors, Semantic errors.

UNIT V

Code Optimization and Code Generation: Local optimization, Loop optimization, Peephole optimization, Basic blocks and flow graphs, DAG, Data flow analyzer, Machine Model, Order of evaluation, Register allocation and code selection.

TEXT BOOKS

1. Introduction to Theory of Computation. Sipser, 2nd Edition,Thomson.
2. Hopcroft, Ullman, "Introduction to Automata Theory, Languages and Computation", Pearson Education
3. Compilers Principles, Techniques and Tools Aho, Ullman, Sethi,Pearson Education

REFERENCES

1. Modern Compiler Construction in C , Andrew W.Appel Cambridge University Press.
2. Compiler Construction, LOUDEN, Thomson.
3. Elements of Compiler Design, A. Meduna, Auerbach Publications, Taylor and Francis Group.
4. Principles of Compiler Design, V.Raghavan, TMH.
5. Engineering a Compiler, K. D. Cooper, L.Torczon, ELSEVIER.
6. Introduction to Formal Languages and Automata Theory and Computation - KamalaKrithivasan and Rama R, Pearson.

GOKARAJU RANGARAJU INSTITUTE OF ENGINEERING AND TECHNOLOGY

MACHINE LEARNING

Course Code:GR22A3140 **L/T/P/C: 3/0/0/3**

III Year II Semester

Prerequisites:

1. Mastery of introduction-level algebra, statistics, and probability theory
2. Fundamental knowledge on data modeling and evaluation

Course Outcomes:

1. Explain the concepts machine learning models and able to perform feature engineering techniques.
2. Identify and apply appropriate supervised learning models.
3. Design neural network models for the given data.
4. Perform evaluation on machine learning algorithms and model selection techniques.
5. Devise un-supervised and reinforcement learning models.

UNIT I

Introduction: Introduction to machine learning, Supervised learning, Unsupervised learning, Semi-supervised learning, Reinforcement learning, Deep learning, Concept learning using find-S algorithm.

Feature Engineering: Feature Selection using Filter, Wrapper, Embedded methods, and Feature normalization using min-max normalization, z-score normalization, and constant factor normalization,

Introduction to Dimensionality Reduction: Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA) techniques.

UNIT II

Supervised Learning – I (Regression and Classification)

Regression models: Simple linear regression, Multiple linear regression. Cost Function, Gradient Descent, Performance Metrics: Mean Absolute Error (MAE), Mean Squared Error (MSE), R-Squared error, Adjusted R Square.

Classification models: Decision Trees-ID3, CART, Naive Bayes, K-Nearest-Neighbors (KNN), Logistic regression, Multinomial logistic regression, Support Vector Machines (SVM).

UNIT-III

Supervised Learning – II (Classification with Neural Networks)

Neural Network Representation: Introduction to Artificial Neural Networks (ANN), Activation Functions, Perceptron and Back Propagation algorithms.

Convolutional Neural Networks - Convolution and Pooling layers, Recurrent Neural Networks(RNN).

Classification Metrics: Confusion matrix, Precision, Recall, Accuracy, F-Score, ROC curves

UNIT IV

Model Validation in Classification: Cross Validation - Holdout Method, K-Fold, StratifiedK-Fold, Leave-One-Out Cross Validation (LOOC-V).
Bias-Variance tradeoff, Regularization, Overfitting, Underfitting.
Ensemble Methods: Boosting, Bagging, Random Forest classifier.

UNITV

Unsupervised Learning: K-means, K-Modes, K-Prototypes, Gaussian Mixture Model clusteringalgorithms, Expectation-Maximization, BIRCH clustering algorithms.
Reinforcement Learning: Exploration and exploitation trade-offs, non-associative learning, Markov decision processes, Q-learning.

Text books:

1. Data Mining–Concepts and Techniques -Jiawei Han and Micheline Kamber, Morgan Kaufmann (2nd Edition).
2. Machine Learning – Tom M. Mitchell, TMGH.
3. Introduction to Data Mining - Michael Steinbach, Pang-Ning Tan, and Vipin Kumar (Second Edition).

References:

1. Kevin Murphy, Machine Learning: A Probabilistic Perspective, MIT Press, 2012.
2. R. S. Sutton and A. G. Barto. Reinforcement Learning - An Introduction. MIT Press,1998.
3. Trevor Hastie, Robert Tibshirani, Jerome Friedman, The Elements of Statistical Learning, Springer 2009.
4. Christopher Bishop, Pattern Recognition and Machine Learning, Springer, 2007.
5. Machine Learning Yearning, Andrew Ng.

GOKARAJU RANGARAJU INSTITUTE OF ENGINEERING AND TECHNOLOGY
FULL STACK DEVELOPMENT

Course Code: GR22A3123 **L/T/P/C: 3/0/0/3**
III Year II Semester

Course Outcomes:

At the end of this course, students will able to

1. Analyse the Structure and implement HTML/CSS and apply intermediate and advanced web development practices.
2. Implement the basic JavaScript and Develop a fully functioning website and deploy on a web server.
3. Design the applications using bootstrap, jQuery and Angular JS
4. Construct the real-world applications using PHP and MySQL
5. Become an industry-ready engineer who can be readily deployed in a project

UNIT I

Introduction to Full Stack Development & HTML5

Introduction, Modern Application Architecture, Front-End Technologies, Front End Libraries and Frameworks, Back-end Technologies.

HTML 5 - New Elements, Video & Audio, Canvas, Vector Graphics, Web Storage, Drag & Drop, Geolocation.

UNIT II

CSS3 & Java Script

CSS3: Basic Styling, Positioning & Background Images, Pseudo Classes, Colors, Backgrounds & Gradients, Text & Box Shadows, Transitions & Animation, Columns & Flexbox.

Java Script: Variables, Arrays & Objects, Loops, Conditionals & Switches, Functions & Events, JavaScript Form Validation

UNIT III

Bootstrap and React JS

Bootstrap Programming - Setup, Templates & Navbar, Typography, Forms & Tables, CSS Components, Grid System, Modal, Dropdown, Tabs & Tooltip, Collapse, Accordion and Carousel.

React JS - Introduction, Features, Components - Lifecycle, Constructor, Forms, Events.

UNIT IV

jQuery and Angular JS

jQuery - Overview of jQuery, Configuration of jQuery, syntax, selectors, events, effects, working with HTML, jQuery with CSS, Traversing.

Angular JS – Introduction to AngularJS, Features of AngularJS-MVC Architecture and Expressions -Numbers, Strings, Objects, Arrays, Directives, Filters.

UNIT V

PHP and MySQL

Backend Programming with PHP - XAMMP Environment Setup, PHP Programming Fundamentals, PHP Data Types & Dates, Objects and Classes, Cookies, Sessions.

Introduction MySQL - Creating a MySQL Database, Connect & Fetch data from MySQL, Insert and Delete MySQL data from PHP.

Textbooks:

1. Full Stack Web Development For Beginners: Learn Ecommerce Web Development Using HTML5, CSS3, Bootstrap, JavaScript, MySQL, and PHP by Riaz Ahmed – 2021
2. Complete Course Full Stack Web Development – Matthew J. Leonard

References:

1. Modern Full-Stack Development: Using TypeScript, React, Node.js, Webpack, and Docker Paperback – 30 March 2020,by Frank Zammetti (Author)
2. Mastering Html, Css & Javascript Web Publishing Paperback – 15 July 2016 by Laura Lemay (Author), Rafe Colburn (Author), Jennifer Kyrnin (Author)
3. The Full Stack Developer: Your Essential Guide to the Everyday Skills Expected of a Modern Full Stack Web Developer Paperback – Import, 20 November 2018 by Chris Northwood (Author)

GOKARAJU RANGARAJU INSTITUTE OF ENGINEERING AND TECHNOLOGY
BIG DATA ANALYTICS
(PROFESSIONAL ELECTIVE-II)

Course Code: GR22A3143 **L/T/P/C: 3/0/0/3**
III Year II Semester

Pre - Requisites:

Students should have knowledge of one Programming Language (Java/ Python), thorough practice of SQL, and exposure to Linux/ UNIX Environment.

Course Outcomes:
1. Interpret the need for HDFS in processing big data.
2. Examine the purpose of data ingestion tools in Big Data Systems.
3. Implement distributed processing of big data using Map Reduce or Pig.
4. Compose Queries with Hive or HBase to analyze the data.
5. Inspect the components of Apache Spark to perform in memory processing.

UNIT I
Introduction to Big Data and Hadoop:
Types of Digital Data, Definition of Big Data, V's of Big Data, Advantages of Big Data, Characteristics of Hadoop, RDBMS Vs Hadoop, Ecosystem components of Hadoop, Big Data Analytics Pipeline, Hadoop Distributions, Need for HDFS, Characteristics of HDFS, HDFS Components, HDFS High Availability Architecture, Block Replication Method, Rack Awareness, HDFS Commands.

UNIT II
Data Ingestion into Big Data Systems and ETL:
Big Data Ingestion Tools, Apache Sqoop, Benefits of Apache Sqoop, Sqoop Connectors, Importing and Exporting to and from Hadoop using Sqoop, Limitations of Sqoop, Apache Flume Model, Data Sources for FLUME, Components of FLUME Architecture.

UNIT III
Distributed Processing - Map Reduce and PIG:
Need for YARN, Elements of YARN Architecture, Characteristics of Map Reduce, Phases of Map Reduce with an Example, Anatomy of MR Job Run with YARN, Handling Failures, Task Execution, Map Reduce Input and Output Formats, Shuffle and Sort, Joins in Map Reduce, Introduction to PIG, Components of PIG, Data Types in PIG – Simple and Complex, PIG Execution Modes, PIG Interactive Modes, Comparison of PIG with databases, Data Processing Operators.

UNIT IV
Apache Hive and NOSQL Database - HBase:
Features of HIVE, HIVE Architecture, HIVE Metastore, Datatypes in HIVE, HIVEQL, Tables, File Format Types–Text, Sequence, AVRO, Parquet, Querying Data, Types of

NOSQL Database, Characteristics of HBASE, Architecture, HBase Vs RDBMS, HBASE Shell Commands.

UNIT V

Apache Spark:

Functional Programming, Components of Apache Spark, Applications of in - memory processing, Hadoop Ecosystem Vs. Spark, Spark Architecture, RDDs in Spark, SparkSQL, Architecture of SparkSQL, DataFrames, Data Analytics, Types of Analytics.

TEXT BOOKS:

1. Tom White “Hadoop: The Definitive Guide”4thedition, O’reily Media, 2012.
2. Seema Acharya, Subhasini Chellappan, "Big Data Analytics" Wiley 2015.

REFERENCES:

1. MichaelBerthold,DavidJ.Hand,"Intelligent DataAnalysis”,Springer,2007.
2. Jay Liebowitz, “Big Data and Business Analytics” Auerbach Publications, CRCpress(2013)
3. Tom Plunkett, Mark Hornick, “Using R to Unlock the Value of Big Data: BigData Analytics with Oracle REnterprise and Oracle R Connector for Hadoop”,McGraw- Hill/Osborne Media(2013), Oracle press.
4. Anand Rajaramanand Jefrey David Ulman,“ Mining of Massive Datasets”,Cambridge UniversityPress,2012.
5. Bill Franks, “Taming the Big Data Tidal Wave:Finding Opportunities in HugeDataStreamswithAdvancedAnalytics”,JohnWiley&sons,2012.

GOKARAJU RANGARAJU INSTITUTE OF ENGINEERING AND TECHNOLOGY
UNIX PROGRAMMING
(PROFESSIONAL ELECTIVE-II)

Course code GR22A3124 **L/T/P/C: 3/0/0/3**
III Year II Semester

Prerequisites:

1 Knowledge in C Programming
2 Knowledge in Operating System

Course Outcomes

1. Experiment and work with various shell scripts. Demonstrate the usage of various system calls for file and directory management
2. Develop the ability to implement various process and signal handling mechanisms of UNIX
3. Select the appropriate mechanism for working with IPC using pipes, named pipes along with concurrency management
4. Interpret the usage of system calls for SYSTEM-V IPC using programs
5. Develop the ability to implement various client server applications using sockets

UNIT I

Working with the Bourne shell: shell responsibilities, shell meta characters, shell variables, shell commands, control structures, shell script examples.

Unix Files: Unix file structure, low level file access system calls, usage of open, creat, read, write, close, lseek, stat, fstat, umask, dup, dup2. Formatted I/O, streams and file descriptors, file and directory maintenance system calls chmod, chown, unlink, link, symlink, mkdir, rmdir, chdir, getcwd, Directory handling system calls opendir, readdir, closedir, rewinddir, seekdir, telldir.

UNIT II

Unix Process and Signals: process structure, starting new process, waiting for a process, zombie process, process control, system call interface for process management-fork, vfork, exit, wait, waitpid, exec, system, **Signals**- Signal functions, unreliable signals, kill, raise, alarm, pause, abort, sleep functions.

UNIT III

Interprocess Communication Overview: Introduction to IPC, IPC between processes on a single computer system, file and record locking, other unix locking techniques, pipes, FIFOs, streams and messages, namespaces.

UNIT IV

Message Queues-Unix system-V messages, unix kernel support for messages, unix APIs for messages, client/server example. Semaphores-Unix system-V semaphores, unix kernel support for semaphores, unix APIs for semaphores, file locking with semaphores. Shared Memory-

Unix system-V shared memory, unix kernel support for shared memory, unix APIs for shared memory, semaphore and shared memory example.

UNIT V

Sockets: Introduction to Sockets, Socket system calls for connection-oriented protocol and connectionless protocol, example-client/server programs.

TEXT BOOKS

1. Unix the ultimate guide, Sumitabha Das, TMH
2. Unix Network Programming, W.R.Stevens, Pearson/PHI
3. UNIX Network Programming, Vol. I, Sockets API, 2nd Edition. - W.Richard Stevens, Pearson Edn. Asia.

REFERENCE BOOKS:

1. M.G. Venkatesh Murthy: UNIX & Shell Programming, Pearson Education.
2. Richard Blum , Christine Bresnahan : Linux Command Line and Shell Scripting Bible, 2nd Edition, Wiley,2014.

GOKARAJU RANGARAJU INSTITUTE OF ENGINEERING AND TECHNOLOGY
MOBILE APPLICATION DEVELOPMENT
(PROFESSIONAL ELECTIVE-II)

Course code: GR22A3125 **L/T/P/C: 3/0/0/3**
III Year II Semester

Prerequisites
1. Knowledge in core Programming

Course Outcomes: At the end of the course the student will be able to
1. Design simple GUI applications.
2. Apply key technological principles and methods for delivering and maintaining mobile Applications on Android devices.
3. Make use of built-in widgets and components.
4. Design apps which uses the database to store data locally.
5. Develop the apps that make use of interfaces with built-in views.

UNIT I
Java Concepts: OOPs Concepts, Inheritance in detail, Exception handling, Packages & interfaces, Multi-threading, JVM & .jar file extension, SQL-DML and DDL Queries.

UNIT II
Introduction to Android: What is Android? Setting up development environment, Dalvik Virtual Machine & .apk file extension, Fundamentals: Basic Building blocks - Activities, Services, and Broadcast Receivers & Content providers. UI Components - Views & notifications. Components for communication -Intents & Intent Filters, Android API levels (versions & version names).

UNIT III
Application Structure (in detail): Android Manifest.xml, uses-permission & uses-sdk, Resources & R.java, Assets, Layouts & Drawable Resources, Activities and Activity lifecycle, first sample Application.
Emulator-Android Virtual Device: Launching emulator, editing emulator settings, Introduction to DDMS, Basic UI design, Preferences, Menu, Intents- Implicit Intents, Explicit Intents.

UNIT IV
Content Providers: SQLite Programming, SQLite OpenHelper, SQLite Database, Cursor, Reading and updating Contacts, Reading bookmarks, Examples.

UNIT V
Threads: Threads running on UI thread (runOnUiThread), Worker thread, Handlers & Runnable, AsynTask, and Examples.

UI design, Tabs and Tab Activity, Styles & Themes, Examples.

Text Books

1. Android Programming for Beginners - Second Edition: Build in-depth, full-featured Android 9 Pie apps starting from zero programming experience, 2nd Edition by John Horton.
2. Head First Android Development: A Brain-Friendly Guide by Dawn Griffiths.
3. Android Cookbook: Problems and Solutions for Android Developers, Second Edition (Grayscale Indian Edition) by Ian Darwin**.**

Reference Books

1. Reto Meier, "Professional Android 2 Application Development", Wiley India Pvt Ltd.
2. Mark L Murphy, "Beginning Android", Wiley India Pvt Ltd.
3. Android Application Development All in one for Dummies by Barry Burd, Edition.

GOKARAJU RANGARAJU INSTITUTE OF ENGINEERING AND TECHNOLOGY

CLOUD COMPUTING

(PROFESSIONAL ELECTIVE-II)

Course Code:GR22A3126 **L/T/P/C: 3/0/0/3**

III Year II Semester

Prerequisites:

Students are expected to have knowledge on Operating systems, Virtualization and Networking

Course Outcomes:

1. Identify varies features, advantages and challenges of cloud computing, compare their operation, implementation and performance
2. Applying knowledge of Virtualization, Analyze and compare different types of clouds and cloud services.
3. Validating the financial and technological implications in selecting cloud computing paradigm for an organization.
4. Compare operation and economic models of varies trending cloud platforms in IT Industry.
5. To know how to overcome the security challenges and risks involved in the cloud.

UNIT I

Understanding Cloud Computing: Cloud Computing, Introduction to Cloud Computing, Cloud Architecture and Cloud Services (IaaS, PaaS, SaaS) , Cloud models– Public vs Private, Cloud Technologies for Network-Based System , System Models for Distributed and Cloud Computing , NIST Cloud Computing Reference Architecture

UNIT II

Virtualization: Basics of Virtualization, Types of Virtualizations, Implementation Levels of Virtualization, Virtualization Structures, Tools and Mechanisms, Virtualization of CPU, Memory, I/O Devices, Virtual Clusters and Resource management, Virtualization for Data-centre Automation

UNIT III

Cloud Infrastructure: Architectural Design of Compute and Storage Clouds, Layered Cloud Architecture Development, Design Challenges, Inter Cloud Resource Management, Resource Provisioning and Platform Deployment, Global Exchange of Cloud Resources

UNIT IV

Programming Model: Parallel and Distributed Programming Paradigms, Map Reduce, Twister and Iterative Map Reduce, Hadoop Library from Apache, Mapping Applications, Programming Support, Google App Engine, Amazon AWS, Cloud Software Environments, Eucalyptus, Working with EC2 API, Open Nebula, Open Stack, Aneka, CloudSim

UNIT V

Security in the Cloud: Security Overview, Cloud Security Challenges and Risks, Software- as-a Service Security, Security Governance, Risk Management, Security Monitoring, Security Architecture Design, Data Security, Application Security, Virtual Machine Security, Identity Management and Access Control, Autonomic Security

Text Books:

1. George Reese, "Cloud Application Architectures: Building Applications and Infrastructure in the Cloud"O'Reilly
2. Kumar Saurabh, "Cloud Computing, insights into New-Era Infrastructure", Wiley India.
3. RajkumarBuyya, Christian Vecchiola, S.TamaraiSelvi, 'Mastering Cloud Computing", TMGH.

References:

1. Kai Hwang, Geoffrey C Fox, Jack G Dongarra, "Distributed and Cloud Computing, From Parallel Processing to the Internet of Things", Morgan Kaufmann Publishers.
2. John W. Rittinghouse and James F.Ransome, "Cloud Computing: Implementation, Management, and Security", CRC Press.
3. Toby Velte, Anthony Velte, Robert Elsenpeter, "Cloud Computing, A Practical Approach", TMH.
4. Ronald L. Krutz, Russell Dean Vines, "Cloud Security, A comprehensive Guide to Secure Cloud Computing", Wiley, India.
5. Nick Antonopoulos, Cloud computing, Springer Publications.